Project: Worked examples
Title: Psychrometrics
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1. Scope and method

Calculates the state of moist air from dry-bulb and one other property at a stated pressure or altitude. It then applies one air-handling process with its heat and moisture balance: sensible heating or cooling, dehumidification, humidification or mixing.

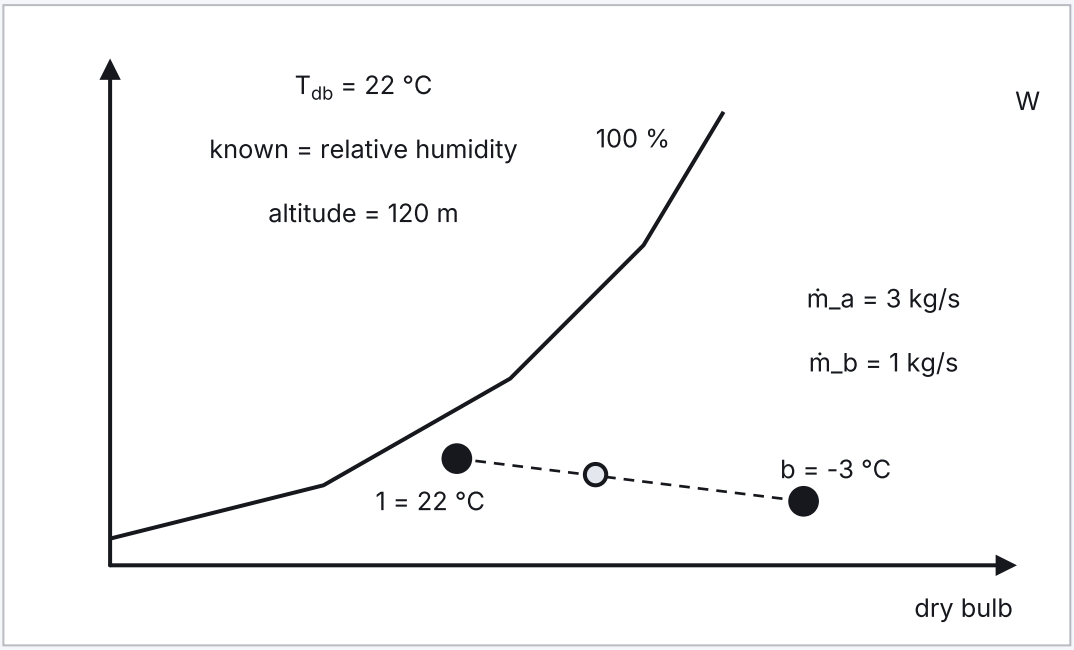


Figure 1 — Adiabatic mixing of two streams: the mixed state lies on the line between them, by mass flow. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Saturation and humidity	$p_{ws}(T)$ by Hyland–Wexler over water at or above 0 °C and over ice below; $W = 0.621945\ p_w/(p - p_w)$; $\varphi = p_w/p_{ws}$
Enthalpy and volume per kilogram of dry air	$h = 1.006\ T + W(2501 + 1.86\ T)$ kJ/kg; $v = 0.287042\ (T + 273.15)(1 + 1.607858\ W)/p$
Dew point and wet-bulb temperatures, by bisection	$p_{ws}(T_{dp}) = p_w$; the wet-bulb from the adiabatic-saturation relation
Sensible heating or cooling at constant moisture content	$\Phi = \dot{m}\ (h_2 - h_1)$
Cooling with dehumidification, less the heat leaving in the condensate	$Q_{total} = \dot{m}(h_1 - h_2) - \dot{m}_w\ h_w$, with $\dot{m}_w = \dot{m}(W_1 - W_2)$
Sensible and latent split of the coil load, along $T_1 \rightarrow T_2$ at W_1 then $W_1 \rightarrow W_2$ at T_2	$Q_{sensible} = \dot{m}(h_1 - h(T_2, W_1))$; $Q_{latent} = \dot{m}(h(T_2, W_1) - h_2)$; $SHR = Q_{sensible}/(Q_{sensible} + Q_{latent})$; with RH_2 blank and T_2 above the entering dew point the coil is dry, $W_2 = W_1$
Steam and adiabatic humidification	$h_2 = h_1 + (W_2 - W_1)\ h_{steam}$ for steam; $h_2 = h_1$ for adiabatic (spray) humidification

WHAT IT CALCULATES	RELATION
Mixing two airstreams, weighted by dry-air mass flow	$W_{mix} = \Sigma \dot{m}\ W/\Sigma \dot{m}$; $h_{mix} = \Sigma \dot{m}\ h/\Sigma \dot{m}$

Basis: Psychrometrics (ideal-gas relations, processes)[1](Ch. 1); Saturation pressure of water and ice[2]; Properties of humid air (principles)[3](§1).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
known	Second known property	relative humidity		assumption: RH given.	any
process	Process	adiabatic mixing		assumption: Mixing box.	any
altitude	Altitude (used when no pressure is entered)	120	m	drawing: Site survey SRV-01: 120 m AOD	-500 ... 5000 m
T_{db}	Dry-bulb temperature	22	°C	spec: Return air 22 °C / 40 %	-40 ... 60 °C
RH	Relative humidity	40		spec: Return air 22 °C / 40 %	5 ... 100
m_{da}	Dry-air mass flow	3	kg/s	spec: AHU-02 return 3 kg/s	0 ... 1000 kg/s
h_{steam}	Steam enthalpy (steam humidification)	2676		assumption: Dry saturated steam at atmospheric pressure, about 2676 kJ/kg from steam tables; used by the steam-humidification run only.	2500 ... 3000
$T_{db,b}$	Second stream dry-bulb (mixing)	-3	°C	spec: CIBSE Guide A winter design −3 °C / 90 %, assumed	-40 ... 60 °C
RH_b	Second stream relative humidity (mixing)	90		spec: Winter design condition	5 ... 100
$m_{da,b}$	Second stream dry-air mass flow (mixing)	1	kg/s	spec: AHU-02 minimum outdoor air 1 kg/s	0 ... 1000 kg/s

3. Assumptions

A1. Moist air is an ideal mixture of ideal gases. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A2. The enhancement factor is taken as 1. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A3. Condensate leaves at the coil leaving dry-bulb. (Process conventions.)
A4. Steam adds the enthalpy h_{steam} at negligible mass of dry air. (Process conventions.)
A5. Adiabatic humidification follows a line of constant enthalpy, which is close to constant wet-bulb. (Evaporative process idealisation.)
A6. known (Second known property) is taken as relative humidity. (RH given.)
A7. process (Process) is taken as adiabatic mixing. (Mixing box.)
A8. h_{steam} (Steam enthalpy (steam humidification)) is taken as 2676. (Dry saturated steam at

atmospheric pressure, about 2676 kJ/kg from steam tables; used by the steam-humidification run only.)

4. Working

[2] p_{atm} = 101.325 (1 - 2.25577e-5 z)^5.2559 [1](Ch. 1 eq. 3)
= z = 120 m
= 99.892 kPa
Barometric pressure (standard atmosphere)

[6] $p_{ws,1}$ = p_{ws}(t) Hyland-Wexler (entering)
= t = 22 °C
= 2.6448 kPa
Saturation pressure, state 1

[7] $p_{w,1}$ = p W/(0.621945 + W)
= W = 0.0066572
= 1.0579 kPa
Vapour pressure, state 1

[8] $T_{1,out}$ = dry-bulb (entering)
= t = 22 °C
= 22 degC
Dry-bulb, state 1

[9] W_1 = 0.621945 p_w/(p - p_w)
= 1.0579 / (99.892 - 1.0579)
= 6.6572
Humidity ratio, state 1 (g/kg dry air)

[10] RH_1 = p_w / p_{ws}
= 1.0579 / 2.6448
= 40
Relative humidity, state 1 (%)

[11] φ_1 = p_w / p_{ws}
= 1.0579 / 2.6448
= 40 %
Relative humidity, state 1, as a percentage quantity (the same value as RH_1; a plain-number output cannot be linked to a percentage input)

[12] $T_{dp,1}$ = p_{ws}(t_{dp}) = p_w, bisection
= p_w = 1.0579 kPa
= 7.7942 degC
Dew point, state 1

[13] $T_{wb,1}$ = adiabatic-saturation relation, bisection
= t = 22, W = 0.0066572
= 13.853 degC
Wet-bulb, state 1

[14] h_1 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 22 + 0.0066572 × (2501 + 1.86 × 22)
= 39.054
Enthalpy, state 1 (kJ/kg dry air)

[15] v_1 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 99.892 kPa
= 0.8572 m³/kg
Specific volume, state 1 (per kg dry air)

[16] ρ_1 = (1 + W)/v
= (1 + 0.0066572) / 0.8572
= 1.1744 kg/m³
Moist-air density, state 1

[22] $p_{ws,b}$ = p_{ws}(t) Hyland-Wexler (second stream)
= t = -3 °C
= 0.47606 kPa

Saturation pressure, state b

[23] $p_{w,b}$ = p W/(0.621945 + W)

= W = 0.0026791

= 0.42845 kPa

Vapour pressure, state b

[24] $T_{b,out}$ = dry-bulb (second stream)

= t = -3 °C

= -3 degC

Dry-bulb, state b

[25] W_b = 0.621945 p_w/(p - p_w)

= 0.42845 / (99.892 - 0.42845)

= 2.6791

Humidity ratio, state b (g/kg dry air)

[26] RH_b = p_w / p_{ws}

= 0.42845 / 0.47606

= 90

Relative humidity, state b (%)

[27] φ_b = p_w / p_{ws}

= 0.42845 / 0.47606

= 90 %

Relative humidity, state b, as a percentage quantity (the same value as RH_b; a plain-number output cannot be linked to a percentage input)

[28] $T_{dp,b}$ = p_{ws}(t_{dp}) = p_w, bisection

= p_w = 0.42845 kPa

= -4.2454 degC

Dew point, state b

[29] $T_{wb,b}$ = adiabatic-saturation relation, bisection

= t = -3, W = 0.0026791

= -3.4949 degC

Wet-bulb, state b

[30] h_b = 1.006 t + W (2501 + 1.86 t)

= 1.006 × -3 + 0.0026791 × (2501 + 1.86 × -3)

= 3.6675

Enthalpy, state b (kJ/kg dry air)

[31] v_b = 0.287042 (t + 273.15)(1 + 1.607858 W)/p

= p = 99.892 kPa

= 0.77963 m³/kg

Specific volume, state b (per kg dry air)

[32] ρ_b = (1 + W)/v

= (1 + 0.0026791) / 0.77963

= 1.2861 kg/m³

Moist-air density, state b

[33] $m_{da,3}$ = ṁ_a + ṁ_b

= 3 + 1

= 4 kg/s

Mixed dry-air mass flow

[34] $p_{ws,2}$ = p_{ws}(t) Hyland-Wexler (mixed)

= t = 15.784 °C

= 1.7935 kPa

Saturation pressure, state 2

[35] $p_{w,2}$ = p W/(0.621945 + W)

= W = 0.0056627

= 0.90129 kPa

Vapour pressure, state 2

[36] $T_{2,out}$ = dry-bulb (mixed)
= t = 15.784 °C
= 15.784 degC
Dry-bulb, state 2
[37] W_2 = 0.621945 $p_w/(p - p_w)$
= 0.90129 / (99.892 - 0.90129)
= 5.6627
Humidity ratio, state 2 (g/kg dry air)
[38] RH_2 = p_w / p_{ws}
= 0.90129 / 1.7935
= 50.253
Relative humidity, state 2 (%)
[39] φ_2 = p_w / p_{ws}
= 0.90129 / 1.7935
= 50.253 %
Relative humidity, state 2, as a percentage quantity (the same value as RH_2; a plain-number output cannot be linked to a percentage input)
[40] $T_{dp,2}$ = $p_{ws}(t_{dp})$ = p_w, bisection
= p_w = 0.90129 kPa
= 5.4665 degC
Dew point, state 2
[41] $T_{wb,2}$ = adiabatic-saturation relation, bisection
= t = 15.784, W = 0.0056627
= 10.314 degC
Wet-bulb, state 2
[42] h_2 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 15.784 + 0.0056627 × (2501 + 1.86 × 15.784)
= 30.207
Enthalpy, state 2 (kJ/kg dry air)
[43] v_2 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 99.892 kPa
= 0.83782 m³/kg
Specific volume, state 2 (per kg dry air)
[44] ρ_2 = (1 + W)/v
= (1 + 0.0056627) / 0.83782
= 1.2003 kg/m³
Moist-air density, state 2

5. Results

SYMBOL	VALUE	DESCRIPTION
$T_{1,out}$	22 °C	Dry-bulb temperature, state 1
φ_1	40 %	Relative humidity, state 1, as a quantity in % (for linking)
$T_{dp,1}$	7.794 °C	Dew point, state 1
$T_{wb,1}$	13.85 °C	Wet-bulb temperature, state 1
W_1	6.6572	Humidity ratio, state 1, in grams per kilogram of dry air
h_1	39.054	Enthalpy, state 1, in kilojoules per kilogram of dry air
$T_{2,out}$	15.78 °C	Dry-bulb temperature, state 2
W_2	5.6627	Humidity ratio, state 2, in grams per kilogram of dry air
h_2	30.207	Enthalpy, state 2, in kilojoules per kilogram of dry air

SYMBOL	VALUE	DESCRIPTION
$p_{ws,1}$	2.645 kPa	Saturation pressure, state 1
$p_{w,1}$	1.058 kPa	Vapour pressure, state 1
RH_1	40	Relative humidity, state 1, as a number from 0 to 100
v_1	0.8572 m³/kg	Specific volume, state 1
ρ_1	1.174 kg/m³	Moist-air density, state 1
$T_{b,out}$	-3 °C	Dry-bulb temperature, state b
$p_{ws,b}$	0.4761 kPa	Saturation pressure, state b
$p_{w,b}$	0.4285 kPa	Vapour pressure, state b
W_b	2.6791	Humidity ratio, state b, in grams per kilogram of dry air
RH_b	90	Relative humidity, state b, as a number from 0 to 100
φ_b	90 %	Relative humidity, state b, as a quantity in % (for linking)
$T_{dp,b}$	-4.245 °C	Dew point, state b
$T_{wb,b}$	-3.495 °C	Wet-bulb temperature, state b
h_b	3.6675	Enthalpy, state b, in kilojoules per kilogram of dry air
v_b	0.7796 m³/kg	Specific volume, state b
ρ_b	1.286 kg/m³	Moist-air density, state b
$p_{ws,2}$	1.794 kPa	Saturation pressure, state 2
$p_{w,2}$	0.9013 kPa	Vapour pressure, state 2
RH_2	50.253	Relative humidity, state 2, as a number from 0 to 100
φ_2	50.25 %	Relative humidity, state 2, as a quantity in % (for linking)
$T_{dp,2}$	5.467 °C	Dew point, state 2
$T_{wb,2}$	10.31 °C	Wet-bulb temperature, state 2
v_2	0.8378 m³/kg	Specific volume, state 2
ρ_2	1.200 kg/m³	Moist-air density, state 2
$m_{da,3}$	4 kg/s	Mixed mass flow

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

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9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Hyland RW, Wexler A. Formulations for the thermodynamic properties of the saturated phases of H2O from 173.15 K to 473.15 K. ASHRAE Trans. 1983;89(2A):500-519.

3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.